

A NORM-ATTAINING DISC-ALGEBRA OPERATOR WITHOUT BOUNDARY EQUICONTINUITY

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SOURCE AND STATUS

This packet concerns arXiv:2305.00859, Bala–Sarkar–Sensarma, *A Bishop-Phelps-Bollobas theorem for disc algebra* [1]. After Corollary 2.2, the source paper says that the authors do not know how to remove the equicontinuity assumption from the preceding theorem, and adds that it is unlikely that equicontinuity is necessary for norm attainment, but that the paper has no evidence for that claim.

Corollary 2.2. *Let $X \subset \mathcal{B}(A(\mathbb{D}))$, $J_0 \subset A(\mathbb{D})$, $\|x\| = 1$, $\epsilon \in (0, 1)$, with $\epsilon \leq \epsilon_0$ some number. Suppose*

$$\mathcal{F} = \{\widetilde{T}g \mid g \in S_{A(\mathbb{D})}\},$$

is equicontinuous at $e^{i\theta_0}$, and

$$\left| \widetilde{T}f_0(e^{i\theta_0}) \right| > 1 - \frac{\epsilon}{3}.$$

Then there exist $N \in S_{\mathcal{B}(A(\mathbb{D}))}$ and $g_0 \in S_{A(\mathbb{D})}$ such that

- (1) $\|Ng_0\| = 1$,
- (2) $\|f_0 - g_0\| < \sqrt{2\epsilon}$, and
- (3) $\|T - N\| < 8\epsilon$.

We do not know how to remove the equicontinuity assumption from the preceding theorem. Furthermore, it is improbable that the equicontinuity assumption is necessary for norm attainment, but we have no evidence to support this claim either. In the following section, we will demonstrate the existence of operators that satisfy the equicontinuity condition.

The result below is a scoped counterexample: it answers only the norm-attainment necessity comment. It does not remove the equicontinuity hypothesis from the Bishop-Phelps-Bollobas approximation theorem.

DEFINITIONS

The disc algebra $A(\mathbb{D})$ is the Banach algebra of functions continuous on $\overline{\mathbb{D}}$ and holomorphic on $\mathbb{D}$, equipped with the supremum norm. For a bounded operator $T : X \rightarrow A(\mathbb{D})$, the source paper asks for equicontinuity at a boundary point $\zeta \in \mathbb{T}$ of the family

$$\{\widetilde{T}x : x \in S_X\} \subset C(\overline{\mathbb{D}}).$$

PROOF INTUITION

The obstruction is already present in the unit ball of the disc algebra. The monomials z^n have norm one and oscillate faster and faster near every boundary point. The identity operator therefore attains its norm at the constant function 1, while its image of the unit sphere contains a sequence that forbids equicontinuity at any boundary point.

THEOREM

There is a norm-attaining operator $T : A(\mathbb{D}) \rightarrow A(\mathbb{D})$ such that $\{\widetilde{T}f : f \in S_{A(\mathbb{D})}\}$ is not equicontinuous at any point of $\mathbb{T}$.

Proof. Let $T = I_{A(\mathbb{D})}$ be the identity operator on $A(\mathbb{D})$. Then $\|T\| = 1$, and T attains its norm at the constant function 1, since $\|T1\|_\infty = 1$.

Fix an arbitrary boundary point $\zeta \in \mathbb{T}$. For each $n \geq 1$, define

$$f_n(z) = \bar{\zeta}^n z^n \quad (z \in \overline{\mathbb{D}}).$$

Then $f_n \in S_{A(\mathbb{D})}$ and $f_n(\zeta) = 1$. Put

$$z_n = \zeta e^{i\pi/n}.$$

Then $z_n \in \mathbb{T}$, $z_n \rightarrow \zeta$, and

$$f_n(z_n) = \bar{\zeta}^n (\zeta e^{i\pi/n})^n = e^{i\pi} = -1.$$

Thus

$$|f_n(z_n) - f_n(\zeta)| = 2$$

while $z_n \rightarrow \zeta$. Consequently the family $S_{A(\mathbb{D})} = I_{A(\mathbb{D})}(S_{A(\mathbb{D})})$ is not equicontinuous at ζ . Since ζ was arbitrary, equicontinuity fails at every boundary point. $\square$

VERIFICATION NOTES AND SCOPE

The proof is analytic and uses no computation beyond rendering the source crop. The counterexample is intentionally narrow. It proves that boundary equicontinuity is not a necessary condition for norm attainment of an operator into $A(\mathbb{D})$. It does not prove the full Bishop-Phelps-Bollobas conclusion for arbitrary operators into $A(\mathbb{D})$, nor does it show that the source theorem's equicontinuity hypothesis can be removed.

Cheap run-index search found no prior packet for arXiv:2305.00859. Exact web searches for the source title together with *disc algebra*, *equicontinuity*, *norm attaining*, and *Bishop-Phelps-Bollobas* did not surface a later settlement or this exact observation. Because the example is elementary, the main human-review question is whether the packet's scope matches the source sentence.

HUMAN REVIEW RECOMMENDATION

Review as a scoped counterexample to the source paper's norm-attainment necessity comment. Do not count it as a solution of the stronger open problem of removing equicontinuity from the BPB theorem.

REFERENCES

- [1] N. Bala, J. Sarkar, and A. Sensarma, *A Bishop-Phelps-Bollobas theorem for disc algebra*, arXiv:2305.00859.